

Chapter 3 – The Interaction

3.1 Introduction

Interaction = **communication between user and computer to perform tasks.**

- **Batch processing:** all input given at once.
- **Highly interactive systems:** continuous input and feedback.

3.2 Models of Interaction

Interaction always involves at least **two participants**: the **user** and the **system**. Both are complex and they **communicate and view the domain and task very differently**. The **interface** must therefore **translate effectively** between them. This translation can fail at many points and for many reasons.

Purpose

- **Understand what happens during interaction.**
- **Identify causes** of interaction problems.
- **Compare** different interaction styles.

Key Terms

- **Domain:** Area of knowledge/work (e.g., graphic design).
- **Task:** Actions performed in a domain.
- **Goal:** Desired outcome.
- **Intention:** Specific action to achieve a goal.
- **Task Analysis:** Study of goals, tasks, intentions, and domain.
- **Task Language:** User's language.
- **Core Language:** System's language.

3.2.2 The Execution–Evaluation Cycle (**Norman's Model**)

Execution Phase

1. **Establish goal.**
2. **Formulate intention.**
3. **Specify action sequence.**
4. **Execute action.**

Evaluation Phase

5. **Perceive system state.**
6. **Interpret system state.**
7. **Evaluate outcome against goal.**

goal not achieved? cycle repeats.

> 'gulfs' arise because user and system do not use the same language (task language vs core language):

Gulf of Execution

user's formulation of actions
≠ actions allowed by the system

- Difference between user's intended actions and available system actions.
- **Smaller gulf = easier interaction**

Gulf of Evaluation

user's expectation of changed system state
≠ actual presentation of this state

- Difference between system output and user's understanding.
- **Smaller gulf = easier interpretation.**

Human Errors

Slip

- Correct goal and action, **wrong execution**.
- Example: Typing mistake.

Mistake

- **Wrong goal or understanding**.
- Example: Misinterpreting an icon.

Worked example — switching on a light

Sitting reading as evening falls, you decide you need more light (goal). You form the intention to switch on the desk lamp, and specify the actions (reach over, press the switch). If someone is closer, the intention/actions may differ (ask them to switch it on) although the goal is identical. After executing, you perceive whether the light is on, interpret the result (if not on: bulb blown? not plugged in? — leading to new goals), and evaluate against the original goal (is there now enough light?). If yes, the cycle is complete; if not, you may form a new intention (turn on the ceiling light too).

3.2.3 The Interaction Framework

› Proposed by **Abowd and Beale**

4 Components

- **User (U)** → Task language
- **System (S)** → Core language
- **Input (I)** → Input language
- **Output (O)** → Output language

Translation	From → To	What it covers / how it is judged
Articulation	User → Input	Expressing intentions through input.
Performance	Input → System	Input translated into system actions.
Presentation	System → Output	System presents results.
Observation	Output → User	User interprets output.

Evaluation Criteria

- Coverage
- Ease of translation
- Expressiveness
- Observation quality

3.3 Frameworks and HCI

- **Ergonomics** → User side.
- **Dialog Design** → Input side.
- **Presentation & Screen Design** → Output side.
- **Social & Organizational Context** → Surrounds entire interaction

3.4 Ergonomics

- Controls should be **organized for quick access**.
- **designing** and **arranging** environments, products, and systems to **fit the people who use them**
- **Ergonomics** (**human factors**)

Organization Types

- **Functional**: **Related controls together**.
- **Sequential**: **Based on order of use**.
- **Frequency**: **Most-used controls easiest to access**.

Guidelines

- Easy reach of controls.
- Important displays at eye level.
- Avoid glare and reflections.
- Provide enough spacing.

Industrial Interfaces

› Glass Interface

glass interface means controlling machines through a computer screen instead of physical dials, switches, and knobs.

- **Computer-screen-based controls.**
- Flexible and cheaper.
- Can display information in multiple ways.

› Problems

- Loss of physical context.
- Easy to get lost in menus.
- Screen limitations.

Indirect Manipulation

Needs:

- Immediate feedback for actions.
- Feedback about actual equipment state.
- Awareness of delays and outdated data.

3.4.2 The Physical Environment of the Interaction

Consider:

- User size variations.

- Visibility of displays.
- Comfortable seating/posture.
- Space for movement.

3.4.3 Health Issues

Physical Position

- Reach controls comfortably.
- Proper support for sitting.

Temperature

- Extreme heat/cold reduces performance.

Lighting

- Avoid eyestrain and glare.

Noise

- Excessive noise harms performance and health.

Time

- Excessive computer use may be harmful.

3.4.4 The Use of Color

Guidelines

- Use distinct colors.
- Avoid blue for critical information.
- Never rely only on color.
- Follow conventions:
 - Red = danger

- **Green = normal**
- **Yellow = standby**
- Color meanings vary across cultures.

3.4.5 Ergonomics and HCI

- Provides design constraints and standards.
- Forms basis for hardware design standards.

3.5 Interaction Styles

1. Command Line Interface (CLI)

Advantages

- **Powerful.**
- **Flexible.**
- **Supports repetitive tasks.**
- **Direct access to functions.**

Disadvantages

- **Commands** must be **memorized.**
- **Difficult** for **beginners.**

2. Menus

- **Options** displayed **on screen.**
- Based on **recognition rather than recall.**
- Often **hierarchical.**
- Easy to learn.
- Suitable for novices.

3. Natural Language

- Uses **everyday language**.

Problems

- **Syntactic ambiguity**.
- **Semantic ambiguity**.
- Pronoun ambiguity.
- **Difficult for computers to understand**.

4. Question/Answer & Query Dialog

Question/Answer

- **User guided through questions**.
- Easy to learn.
- Limited flexibility.

Query Languages

- Used for **database searches**
- **Require specific syntax**.
- **More complex for advanced queries**.

5. Form-Fills & Spreadsheets

Form-Fills

- Similar to paper forms.
- **Easy for data entry and retrieval**.

Spreadsheets

- **Grid of cells** with **values/formulas**.

- Automatic recalculation.
- Input and output are closely integrated.

6. WIMP Interface

WIMP =

- Windows
- Icons
- Menus
- Pointers

Features

- Most common interface style.
- Used in Windows, MacOS, UNIX systems.
- Supports **multiple interaction objects**.

7. Point-and-Click

- User **clicks objects to perform actions**.
- Common in web browsers and multimedia.
- Often used with **touchscreens**

8. Three-Dimensional Interfaces

- **3D-looking controls**.
- 3D workspaces.
- Full **virtual reality**.

Benefits

- More **natural interaction**.
- Uses real-world skills.

Design Guidelines

- Provide landmarks.
- Easy navigation.
- Support "back" movement.
- Use ground plane in VR.

3.6 Elements of the WIMP Interface

Windows

- **Independent screen areas.**
- Can contain text/graphics.

Types

- **Overlapping**
- **Tiled**
- **Cascading**

Components

- **Scrollbars**
 - **Title bars**
 - **Nested windows**
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Icons

- **Small pictures representing objects/windows.**
- **Save screen space.**
- **Allow quick reopening of windows.**

Pointers

- [Cursor controlled by mouse](#) or similar device.

Uses

- [Select objects.](#)
 - Show system mode.
 - Show busy state.
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Menus (WIMP)

Types

- Menu Bar
- Pull-down → mouse hold and drag down menu
- Fall-down → mouse just moves over bar!
- Pop-up
- Pin-up
- Pie Menu

Design Rules

- [Meaningful labels.](#)
- Logical [grouping.](#)
- [Consistent](#) structure.
- [Frequently used items easy to access.](#)
- [Keep opposite actions apart.](#)

Keyboard Accelerators

- [Shortcut keys for menu commands.](#)
 - Improve expert performance.
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Buttons

- **Toggle Buttons:** [Switch between two states.](#)
 - **Radio Buttons:** [Only one option selected.](#)
 - **Check Boxes:** [Multiple options selected simultaneously.](#)
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Toolbars

- [Collection of shortcut icons.](#)
 - [Faster access to common functions.](#)
 - May be [customizable](#).
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Palettes

- Show [available modes and current mode.](#)
 - Help users avoid mode confusion.
 - Examples: color palettes, drawing tools.
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Dialog Boxes

- [Temporary windows.](#)
- Display [warnings, errors, or additional tasks.](#)
- Disappear when task is completed.

3.7 Interactivity

- Defines interactive systems.
- Affects overall user experience.

User Initiative

- Modern systems **allow users to control interaction.**
- Users choose actions freely.

Pre-emptive Elements

- **System temporarily takes control.**

Example: Modal dialog boxes.

Error Handling

- Errors should be detectable.
- Users should be able to recover easily.

3.8 The Context of the Interaction

Social & Organizational Factors

- Interaction affected by **workplace and social environment.**
- Designers must **understand real work practices.**

Presence of Others

Can Improve

- Competition.
- Desire to impress.

Can Reduce Performance

- Learning new skills due to fear of failure.

Motivation Sources

- Fear
- Ambition
- Loyalty
- Self-satisfaction

User Involvement

- **Systems should match actual work requirements.**
- Involve users in design process.

Feedback

- Essential for motivation.
- **Confirms actions and system status.**

3.9 Experience, Engagement and Fun

Users must want to use the system, not just be able to use it.

Flow (Csikszentmihalyi)

- **State of complete engagement.**
- Occurs when challenge matches skill level.

Zone of Proximal Development

- Learning happens just beyond current ability with support.

3.9.2 Designing Experience

- Focus on **underlying experience**, not just functionality.
- Recreate important feelings and engagement.

3.9.3 Physical Design and Engagement

Design Constraints

- **Ergonomic** — buttons can't be pushed if too small or too close.
- **Physical** — device size/nature forces certain control positions/styles
- **Legal and safety** — cooker controls must be far enough from pans to avoid burns, yet high enough to keep small children away.
- **Context and environment** — the microwave's controls are smooth to be easy to clean in the kitchen.
- **Aesthetic** — controls must look good.
- **Economic** — it must not cost too much.

Fluidity

- **Physical actions** should **naturally match logical actions**.
- Related to affordances.

State-Revealing Controls

- **Controls should show current state.**
- Example: switch position indicates ON/OFF.

3.9.4 Managing Value

User Value:

- People use systems when **perceived benefits exceed costs**.

Costs

- Purchase cost.
- Learning effort.
- Time investment.

Design Goal

- **Deliver value quickly.**
- Make **future benefits visible** and **meaningful**.